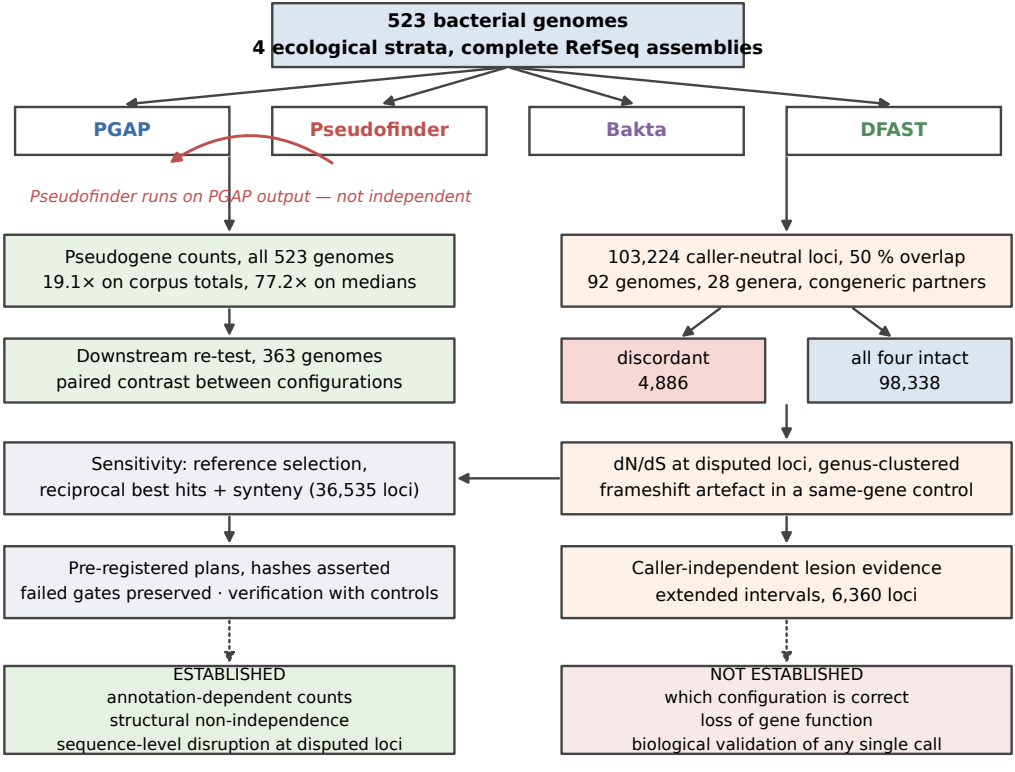
523 bacterial genomes
4 ecological strata, complete RefSeq assemblies
PGAP
Pseudofinder
Bakta
DFAST
Pseudofinder runs on PGAP output — not independent
Pseudogene counts, all 523 genomes
19.1× on corpus totals, 77.2× on medians
103,224 caller-neutral loci, 50 % overlap
92 genomes, 28 genera, congeneric partners
Downstream re-test, 363 genomes
paired contrast between configurations
discordant
4,886
all four intact
98,338
Sensitivity: reference selection,
reciprocal best hits + synteny (36,535 loci)
dN/dS at disputed loci, genus-clustered
frameshift artefact in a same-gene control
Pre-registered plans, hashes asserted
failed gates preserved · verification with controls
Caller-independent lesion evidence
extended intervals, 6,360 loci
ESTABLISHED
annotation-dependent counts
structural non-independence
sequence-level disruption at disputed loci
NOT ESTABLISHED
which configuration is correct
loss of gene function
biological validation of any single call